

Chapter 16 – Event Viewer, Logs Analysis & In-Place OS Repair

1. What is Event Viewer?

Event Viewer is a **powerful Windows tool** used to monitor and analyze everything happening inside the system.

How to Open Event Viewer?

- Press **Windows + R** → type **eventvwr.msc** → Enter
 - OR search **Event Viewer** from Start Menu 
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2. What Does Event Viewer Show?

It records **every important event** in the system:

-  System Logs
 -  Errors
 -  Warnings
 -  Security Logs
 -  Information logs
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3. Why is Event Viewer Important?

Used For:

- Troubleshooting crashes 
 - Finding malware activity 
 - Tracking login attempts 
 - Investigating system issues (Forensics) 
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4. How Event Viewer Works (Concept)

 It acts like a **CCTV camera of your system** 

Everything is recorded:

- Login / Logout
- Software install/uninstall
- Errors & crashes
- Security events

5. Types of Logs (Very Important)

Type	Meaning
 Critical	System crash / sudden shutdown
 Error	Something failed
 Warning	Future problem possible
 Information	Normal activity
 Verbose	Detailed debugging logs

6. Event Viewer Log File Location

 **Path:**

C:\Windows\System32\winevt\Logs

 **File Type:**

.evtx

 **Why Important?**

- Used in:
 - Troubleshooting
 - Digital Forensics

- Security Investigations

👉 Stored in **System32** because it contains **critical security data** 🔒

🧩 7. Application Logs

📍 **Location:**

Event Viewer → Windows Logs → Application

📌 **What it Contains:**

- App install / uninstall
- Updates
- Crashes
- Corruption issues

💡 **Example (Real Scenario):**

👉 User says:

“Antivirus tha phir bhi attack hua”

🔍 **What You Do:**

1. Check Application Logs
2. Find:
 - When antivirus stopped ❌
 - When suspicious software installed ⚠️
3. Correlate with:
 - Temp folder
 - AppData folder

👉 MSI Installer logs show:

- Installation events 📦
-



8. System Logs



Location:

Event Viewer → Windows Logs → System



Use:

- OS-related issues
- Start Menu not working
- System crashes



9. Security Logs (Most Important for Cyber Security)



Location:

Event Viewer → Windows Logs → Security



Shows:

- Login / Logout
- Failed login attempts
- Account changes



10. Important Event IDs (Must Remember)

Event ID Meaning

4624		Successful Login
4625		Failed Login
4720		New User Created
4722		User Enabled
4723		Password Change Attempt

Event ID Meaning

4725  Account Disabled

4726  Account Deleted

Real Interview Scenario:

 If:

- User login successful for months
- Suddenly multiple **4625 (failed logins)**

 Meaning:

- Possible **Brute Force Attack** 
 - Someone trying wrong passwords
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11. Application & Service Logs

 Path:

Event Viewer → Applications and Services Logs → Microsoft → Windows

 Use:

- Find logs for:
 - Windows Defender 
 - System services

Filter Logs:

1. Click **Filter Current Log**
2. Select:
 - Critical 
 - Error 
3. Click OK

👉 Only important issues will show

⚠️ 12. Vulnerability vs Antivirus

Concept	Meaning
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Vulnerability	Weakness in system ⚠️
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Patch	Fix for vulnerability 💊
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Antivirus	Protects from virus 🦠
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👉 Antivirus ≠ Vulnerability Fix

🔄 13. In-Place OS Repair (Very Important) 🛠️

📌 What is it?

Repair Windows **without deleting data**

👉 Used when:

- OS corrupted
 - System unstable
 - Files damaged
-

14. In-Place OS Repair – Step by Step

✓ Step 1:

- Download Windows ISO (same or newer version)

✓ Step 2:

- Open ISO → run:

setup.exe

✓ Step 3:

- Uncheck:

👉 “I want to help make installation better”

✓ Step 4:

- Click:

👉 Change how setup downloads updates

👉 Select:

Not Right Now ❌

✓ Step 5:

- Accept License Agreement ✓

✓ Step 6:

- Choose:

👉 **Keep personal files and apps** 📁

✓ Step 7:

- Click Install 🚀

🕒 Takes: **30–35 minutes**

⚠ Important Notes:

- System may **pause at 83% & 91%** → Don't cancel ❌
 - It's normal behavior
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15. Common Errors in In-Place Repair



Problem 1:

- OS installed in different drive (not C:)



Problem 2:

- Using older ISO than installed OS



Solution:

- Use **same or newer version ISO**
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16. Why Logs Stored in System32?



Because:

- Contains sensitive data:
 - Security logs 🔒
 - Password attempts
 - Defender logs



System32 = **Highly protected folder**



17. Extra Knowledge (Career Tip)

- Learn both:
 - Windows 💻
 - macOS 🍏

macOS Terminology:

Windows

macOS

Control Panel

System Preferences

Command Prompt Terminal

Task Manager

Activity Monitor

 Many cybersecurity companies prefer **macOS environments**

18. Final IT Support Tips

- ✓ Always check Event Viewer first 
 - ✓ Correlate logs with real issue
 - ✓ Focus on:
 - Security logs
 - Application logs
 - ✓ Use filtering to save time 
 - ✓ Think like investigator 
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Final Summary

- Event Viewer = **System CCTV** 
 - Logs = **Proof of activity** 
 - Event IDs = **Investigation tools** 
 - In-place repair = **Safe OS recovery** 
-

Q & A

SECTION 1: BASIC / THEORY QUESTIONS

? 1. What is Event Viewer?

✓ Answer:

Event Viewer is a built-in Windows tool used to **monitor, analyze, and troubleshoot system activities** by viewing logs.

👉 It records:

- System events 
- Errors 
- Security logs 
- Application activity 

💡 Simple Explanation:

Event Viewer works like a **CCTV camera**  that records everything happening inside the system.

? 2. How do you open Event Viewer?

✓ Answer:

There are multiple methods:

1 Press Windows + R → type:

eventvwr.msc

2 Search “Event Viewer” in Start Menu 

? 3. What types of logs are available in Event Viewer?

✓ Answer:

Type	Meaning
✗ Critical	System crash / shutdown
⚠ Error	Something failed
⚠ Warning	Future issue possible
i Information	Normal activity
🔍 Verbose	Detailed debug logs
👉	These logs help identify severity of issues

? 4. What are the main log categories?

✓ Answer:

1 Application Logs 📦

- App crashes
- Installation/uninstallation

2 System Logs 💻

- OS errors
- Driver issues

3 Security Logs 🔒

- Login attempts
 - Account activity
-

? 5. Where are Event Viewer logs stored?

✓ Answer:

📍 Path:

C:\Windows\System32\winevt\Logs

📄 File Type:

- .evtx

💡 Stored in **System32** because:

- Contains sensitive security data 🗝️

? 6. What is the importance of Event Viewer in IT Support?

✓ Answer:

Event Viewer is used for:

- ✓ Troubleshooting system crashes 🌟
- ✓ Investigating malware 🦟
- ✓ Monitoring login activity 🗝️
- ✓ Performing digital forensics 🕵️

🗝️ SECTION 2: SECURITY & EVENT ID QUESTIONS

? 7. What are important Event IDs you must know?

✓ Answer:

Event ID Meaning

4624 Successful login ✓

4625 Failed login ✗

4720 New user created 👤

Event ID Meaning

- 4722 User enabled 
- 4723 Password change attempt 
- 4725 Account disabled 
- 4726 Account deleted 

? 8. How do you detect a brute force attack using Event Viewer?

✓ Answer:

🔍 Steps:

- 1 Open **Security Logs**
- 2 Filter Event ID **4625**
- 3 Look for:
 - Multiple failed login attempts

💡 Conclusion:

👉 Repeated 4625 logs = possible **brute force attack** ⚠️

? 9. What is the difference between vulnerability and antivirus?

✓ Answer:

Concept	Meaning
Vulnerability 	Weakness in system
Patch 	Fix for vulnerability
Antivirus 	Detects/removes malware

👉 Important:

Antivirus ≠ vulnerability fix

SECTION 3: PRACTICAL / REAL-LIFE QUESTIONS

? 10. A user says “System crashed suddenly.” How will you analyze using Event Viewer?

✓ Answer (Step-by-Step):

1 Open Event Viewer

2 Go to:

 Windows Logs → System

3 Filter:

- Critical 
- Error 

4 Check:

- Time of crash
- Event description

5 Identify:

- Driver issue
- Power failure
- Hardware issue

🎯 Conclusion:

Event Viewer helps find **root cause of crash**

? 11. How do you check if antivirus stopped working before an attack?

✓ Answer:

1 Go to:

👉 Application Logs

2 Look for:

- Antivirus service stopped ✗
- Error logs

3 Check installation logs (MSI Installer)

4 Correlate with:

- AppData
- Temp folder

🎯 Conclusion:

You can prove **when antivirus stopped + when attack happened**

? 12. How to filter important logs in Event Viewer?

✓ Answer:

🔧 Steps:

1 Select log (System/Application)

2 Click:

👉 Filter Current Log

3 Select:

- Critical ✗
- Error ⚠

4 Click OK

🎯 Shows only important issues → saves time ⌚

? 13. How do you investigate login activity?

✓ Answer:

1 Go to:

👉 Security Logs

2 Check:

- 4624 → Successful login
- 4625 → Failed login

3 Analyze:

- Time
- Username
- Frequency

🎯 Used in:

- Security monitoring
- Incident response

🔧 SECTION 4: IN-PLACE OS REPAIR QUESTIONS

? 14. What is In-Place OS Repair?

✓ Answer:

In-Place OS Repair is a method to **repair Windows without deleting data or apps**

👉 Used when:

- OS corrupted
 - System unstable
 - Files damaged
-

? 15. How do you perform In-Place OS Repair?

✓ Answer (Step-by-Step):

1 Download Windows ISO (same/newer version)

2 Mount ISO

3 Run:

setup.exe

4 Select:

- Keep personal files and apps 📁

5 Click Install 🚀

🕒 Time: 30–35 minutes

? 16. What are common issues during In-Place Repair?

✓ Answer:

✗ Using older ISO

✗ OS installed on different drive

👉 Fix:

- Use same or newer ISO
-

? 17. What happens if system freezes during installation (83% / 91%)?

✓ Answer:

👉 It is **normal behavior**

✗ Do NOT cancel

✓ Wait for completion

SECTION 5: ADVANCED TROUBLESHOOTING QUESTIONS

 **18. System is slow and unstable. How will you use Event Viewer?**

 **Answer:**

1 Check System logs

2 Filter errors

3 Identify:

- Driver failures
- Disk errors
- Service failures

4 Take action:

- Update drivers
 - Run disk check
-

 **19. How do you correlate logs with real issue?**

 **Answer:**

 **Match:**

- Time of issue 
- Log timestamp 
- User activity

 **Example:**

- Crash at 5 PM
- Log shows driver error at 5 PM

 **Root cause found** 

? 20. Why should IT support always check Event Viewer first?

✓ Answer:

- ✓ Gives exact error details
 - ✓ Saves troubleshooting time
 - ✓ Provides proof of issue
 - ✓ Helps in root cause analysis
-

SECTION 6: INTERVIEW SCENARIO QUESTIONS

? 21. User says “System restarted automatically.” What will you check?

✓ Answer:

- 1 Open System logs
 - 2 Look for:
 - Critical errors
 - Power failure logs
 - 3 Check:
 - Event ID related to shutdown
-

? 22. User reports suspicious login attempts. What will you do?

✓ Answer:

- 1 Open Security logs
 - 2 Filter:
 - Event ID 4625
 - 3 Analyze frequency
 - 4 Inform security team
-

? 23. Application crashes frequently. What will you check?

✓ Answer:

1 Open Application logs

2 Find:

- Error logs
- Faulting module

3 Reinstall/update app

KNM